

Lane–Emden

SIRPY Problem Library

Luminary Quantum Institute, LLC (dba Mitra Institute of Education)

SIRPY · Problem 18

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

A division by zero in the initial condition

$$y'' = -\frac{2y'}{x} - y^5, \quad y(0) = 1, \quad y'(0) = 0$$

This is the Lane–Emden equation of index 5 — a self-gravitating sphere of gas in equilibrium. A star. It has been in the astrophysics curriculum since the 1870s.

Now look at where the data sits. **At** $x = 0$ — exactly where $2y'/x$ is undefined.

That is not a defect in the model. It is the centre of the star, and the only physically natural place to start. But it means most solvers cannot begin here at all. The standard workaround is to start at $x = 10^{-6}$ instead, with values from an expansion you derive by hand, off to the side, checked by nobody.

What we typed

```
solve_ivp(order = 2, f_str = "-2*y1/x - y**5", x0 = 0,
          ic = {0 = 1, 1 = 0})
```

$x_0 = 0$. The singular point itself. No offset. No expansion. No endpoint. Nothing but the problem.

What came back

```
INFO: x0 = 0 is a REGULAR SINGULAR point: y'' + (2/x)y' + N = 0 with
      N = y**5 analytic. Indicial roots {0, -1}: the analytic branch is
      the UNIQUE solution with y(0) = 1 (the companion behaves like
      x^-1 and is discarded by regularity; y'(0) = 0 is forced and
      matches your data).
```

It recognised the point, chose the physically meaningful solution, and said why the other one is not admissible — a star has no infinite density at its centre.

Then it told us something about **our own input**: “ $y'(0) = 0$ is forced and matches your data.” We supplied that condition. It was never ours to choose. Had we written $y'(0) = 0.3$, we would have been handed that fact in the first line of output instead of a plausible-looking wrong answer.

It knows where its own answer stops being trustworthy

```
INFO: Frobenius START segment on [0, 0.0175412]: ... radius estimate
      1.75. Series-piece relative defect on (0, eps]: 2.50e-16.
      Hand-off OVERLAP certificate: series vs march agree to 5.55e-17
      at x = 0.0351.
```

Two numbers worth pausing on.

5.55×10^{-17} — where the start hands off to the main solution, the two are compared on their overlap and the disagreement is *measured*. Seams are where silent errors live. This one is reported, not assumed away.

Radius estimate 1.75. The nearest trouble spots for this solution sit at $\pm i\sqrt{3} = \pm 1.7320508 i$ — off the real line entirely, invisible to any sampling of real values. SIRPY put them at 1.75. **About one percent, without ever leaving the real axis.**

No endpoint given? It says what it chose

INFO: no x1 was given. sirpy measured the first reliable radius (~0.0138) and marched up to 300 radii, capped at a modest default horizon (target $x = 4.16602$) ... Pass $x1=<value>$ to choose the range yourself.

INFO: verification by substitution - VERIFIED: satisfies the equation to $1.40\text{e-}11$ on $[0.0175412, 4.16602]$. Interface continuity across 22 node(s): max one-sided jump in $(y..y1) = 2.7\text{e-}12$ - measured, not assumed. Also checked: initial data to $0.00\text{e+}00$.

solve time: 0.274 s

We gave no interval. It measured its own reach, stated the rule it applied, named where it stopped, and handed the choice back.

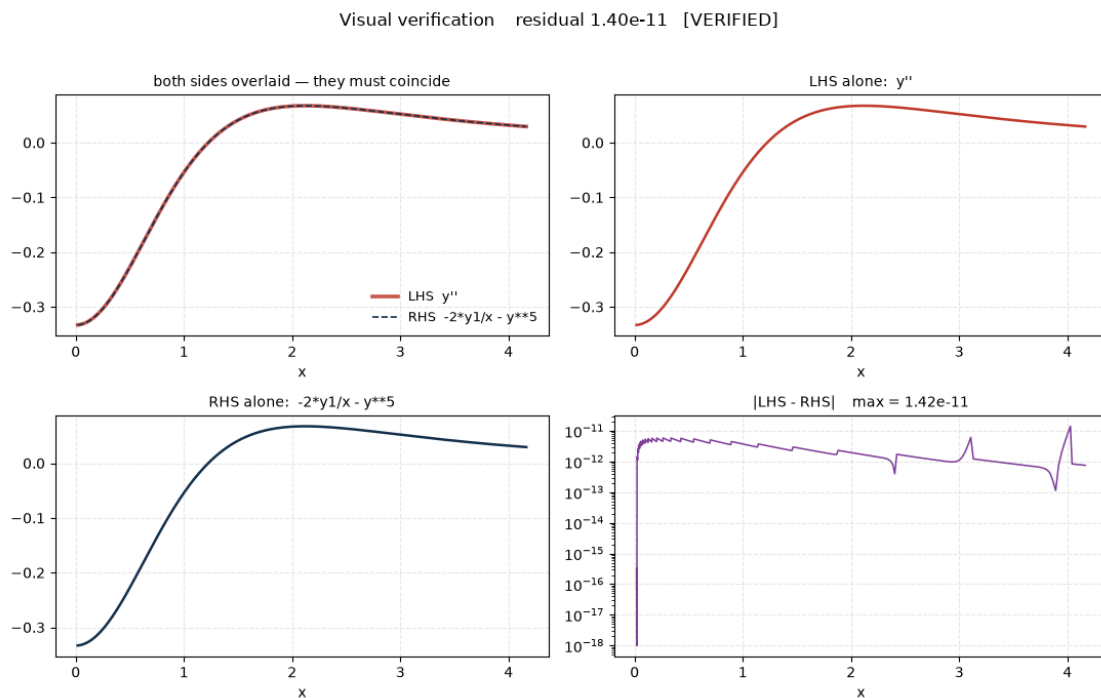


Figure 1: Verification plots.

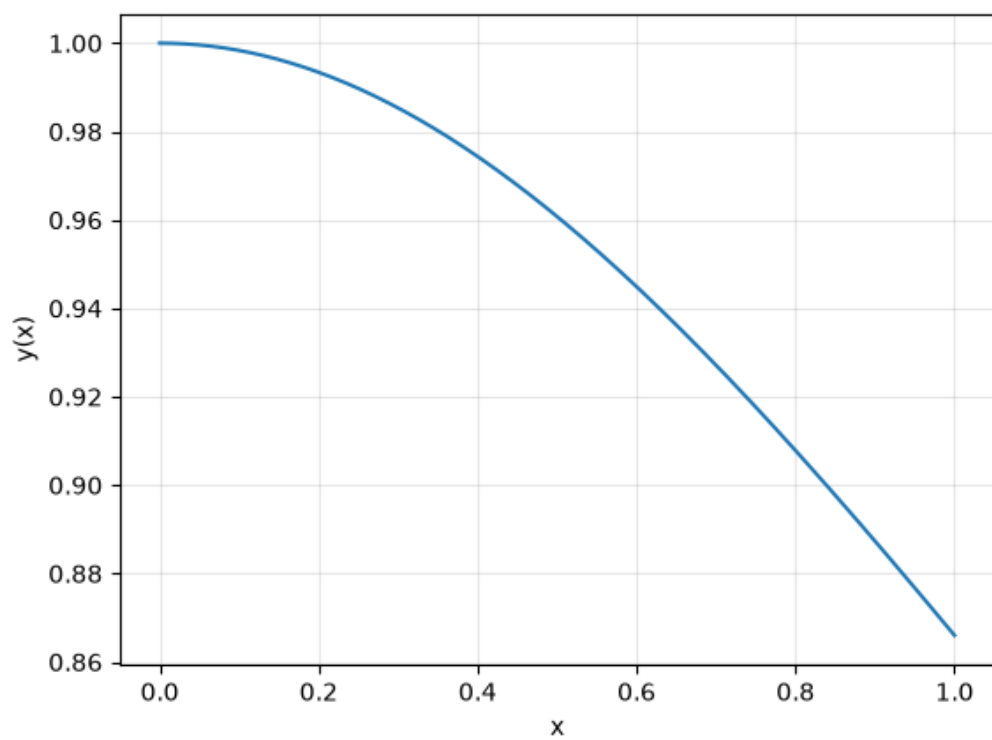


Figure 2: solution plot.

Taking the choice

```
solve_ivp(order = 2, f_str = "-2*y1/x - y**5", x0 = 0,  
          ic = {0 = 1, 1 = 0}, x1 = 100)
```

INFO: the solution was continued analytically over 34 segments.
INFO: verification by substitution - VERIFIED: satisfies the equation to
4.46e-11 on [0.0175412, 100]. Interface continuity across 33
node(s): max one-sided jump in (y..y1) = 6.6e-11 - measured, not
assumed.
solve time: 0.094 s

Twenty-four times the interval. Eleven more segments. Same accuracy. Under a tenth of a second.

Index 5 is the special one: this star has finite mass but *infinite radius* — the density never quite reaches zero, so there is no surface to stop at. The solution simply keeps going, and so does the solver.

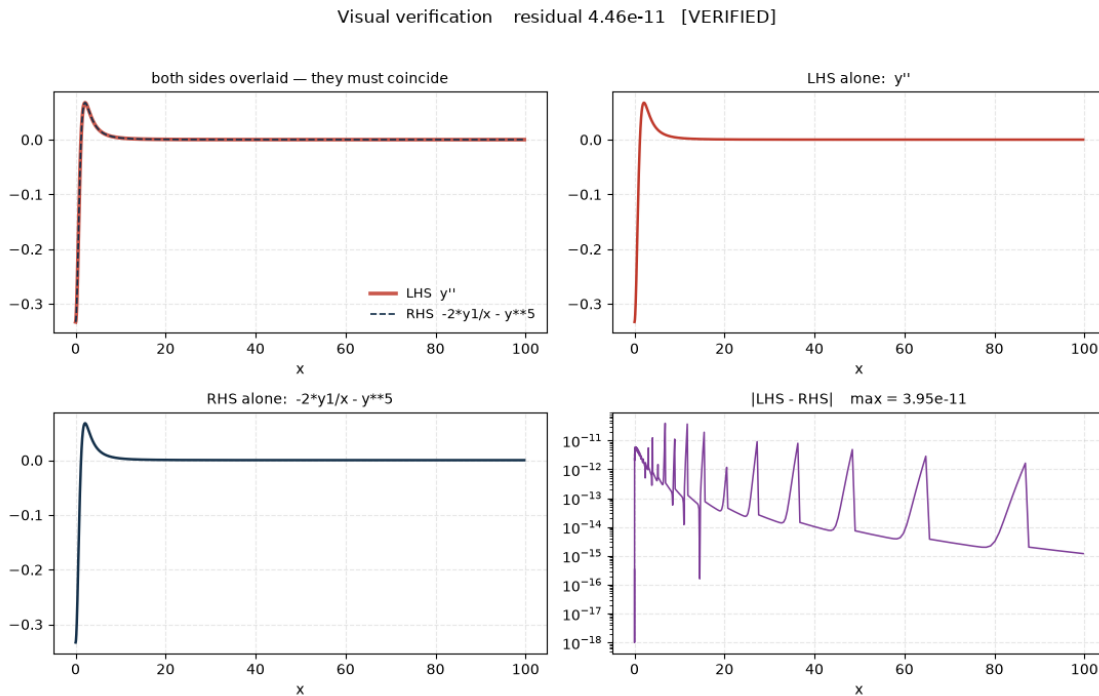


Figure 3: Verification plots: x1 =100

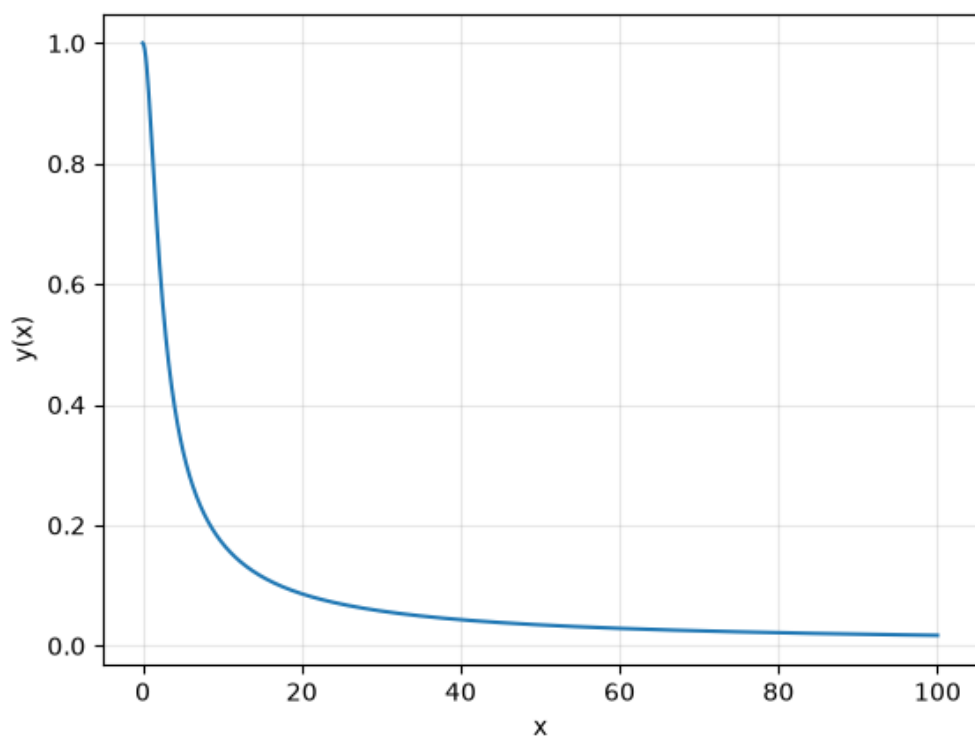


Figure 4: solution plot: $x_1 = 100$

The line we are proudest of

INFO: pole-estimator claims SUPPRESSED at 0.0322433: each lies inside [0.0175412, 100], where verification MEASURED the equation as satisfied. A function cannot satisfy the equation there and be singular there, so the measurement stands and the estimate is discarded.

One of SIRPY's own diagnostics produced a **false alarm** — a singularity at $x = 0.0322$, where the solution is perfectly smooth.

SIRPY caught it, discarded it, and showed the reasoning: the equation had already been *measured* as satisfied there, and nothing can satisfy an equation at a point and be singular at that point.

Most tools would have printed the warning and left you to worry about it. This one printed the argument for ignoring it.

That is the rule the whole library runs on: **measurement outranks estimate — and when they disagree, you are told which won.**

Everything above, in one place

	default run	x1 = 100
interval reached	4.166	100
analytic segments	23	34
equation defect	1.40e-11	4.46e-11
interface agreement	2.7e-12	6.6e-11
initial data	exact	exact
solve time	0.274 s	0.094 s

Three lines of input. Everything else — the singular start, the branch selection, the accuracy, the seams, the false alarm — came back on its own.

Try it. The exact solution here is $y = (1 + x^2/3)^{-1/2}$, so every number above is one you can check yourself.

Contac Us

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SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.

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